

PREVis questionnaire

There are 4 individual scales in PREVis, each measuring a particular dimension of perceived readability.

Understand scale

obvious	It is obvious for me how to read this visualization • can you distinguish the different hydrodynamic attributes ? • can you distinguish between ocean- and terrestrial elements ? • can you geographically locate the area ?
represent	I can easily understand how the data is represented in this visualization • can you tell which attribute is represented by which symbol(s) or colour map (i.e. visual channel) ? • can you identify the baseline (i.e. 0-point) of the attribute from the symbol(s) or colour map ?
understand	I can easily understand this visualization => expected insight <u>just from the visuals</u> (→ how many can you easily identify ?): • the water at the northern coastline of island flows towards the coastline • the water at the southern coastline of island flows away from the coastline • the ocean around the island has a turbulent, non-laminar flow, meaning: the flow direction varies visibly & has no obvious general flow direction • the eddies are areas with 0 m/s velocity at the center, and a divergence of 0 m/s at its boundary • flow direction, divergence and eddy-location are correlated • in the Irish Sea, between the mainland and the island, there are several smaller eddies • there are 2 clockwise- and 1 counter-clockwise turning eddy north of the island • the winding of eddies south of the island varies – some turn clockwise, some turn counter-clockwise • the seawater temperature at the coast differs from than in the (northerly) open ocean • there is a deep ocean trench off the south-southeast center shoreline of the island

Reading data scale

find	I can easily find specific elements in this visualization • can you locate areas of fast-/slow-flowing water ? • can you locate areas of calm, nearly-steady waters ? • can you locate ocean trenches- and ridges ?
identify	I can easily identify relevant information in this visualization • can you identify & distinguish areas of fast-flowing water from areas with slow-flowing water ? • can you identify & distinguish areas of equal flow direction from areas with a different flow direction ? • can you distinguish between warm and cold water areas ? • can you distinguish between deep and shallow waters ?
information	I can easily retrieve information from this visualization • can you directly approximate the flow velocity (in [m/s]) in the visualization ? • can you directly read- or approximate the seabed elevation (i.e. bathymetry; in [m]) from the visualization ? • can you infer the ocean water temperature (in [°C]) from the visualization?

Layout scale

messy	I don't find this visualization messy • can you focus on individual visual elements in the visualization ? • can you distinguish lines from points ? • can you see the background even in densely-sampled areas ? • can you identify all separate colours, or do you get the impression that there are duplicate colours due to colour deficiencies ?
crowd	I don't find this visualization crowded • do you <i>perceive</i> a lot of occlusions in the visualization ? • do you perceive the visualization to be <i>nervous</i> ?
distract	I don't find distracting parts in this visualization • are there graphical elements that have nothing to do with the data, the scale or the orientation ? • If the visualization uses saturation/brightness and transparency: are there discrepancies between the overlay, the data scale and the background colour ?

Reading features scale

visible	I <i>perceive</i> data features (for example, a minimum, or an outlier, or a trend) to be visible in this visualization (i.e. they are <i>visibly existent</i>)
see	I can see data features (for example, a minimum, or an outlier, or a trend) clearly in this visualization (i.e. they are <i>distinct</i> from there surrounding)